

Qatar-1b

R-0476

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180812 · created 2026-07-20T04:16:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458342.8415 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.056
Transit depth (Rp/Rs)^2	2.64 +/- 1.81 [%]
Orbital Inclination (inc)	79.27 +/- 2.92
Airmass coefficient 1 (a1)	0.23 +/- 0.12
Airmass coefficient 2 (a2)	1.19 +/- 0.8
Scatter in the residuals of the lightcurve fit is	48.07 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.024 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.056 vs 0.1463 (deviation 0.28σ)
Duration fit vs published	0.024 d vs 0.0692 d (deviation 1.67σ)
Mid-time O–C	3.0 min
Depth SNR	2.9
Residual scatter	48.07 %

Light curve

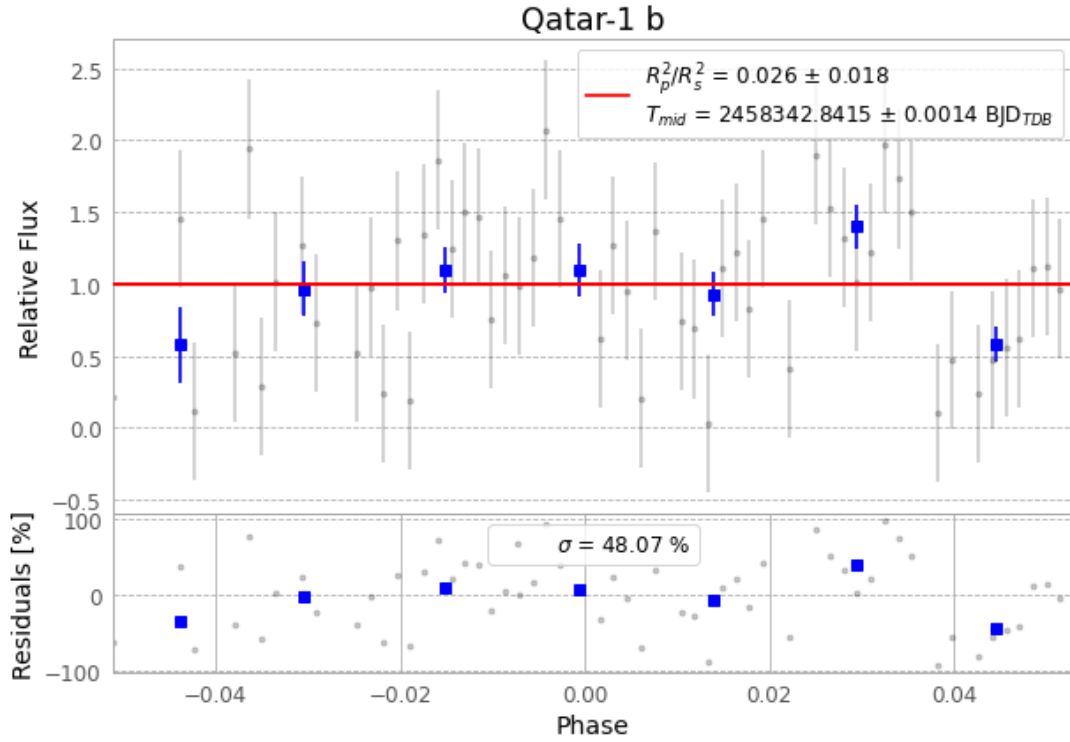


Figure 1 — FinalLightCurve_Qatar-1 b_2018-08-11.png (SHA-256 eecb129e0efcf264...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [94, 409], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fff0f87b4770ca31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

72 FITS frames (47 MB), Qatar-1180812062412.FITS → Qatar-1180812095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180812.log (3439933e6ab5...), FinalLightCurve_Qatar-1 b_2018-08-11.png (eecb129e0efc...), FinalLightCurve_Qatar-1 b_2018-08-11.csv (67dc555731d9...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 04:16Z.